

ZP-DOC-00

Documentation Standard

Rules for identifying, writing, styling, reviewing and controlling ZedPath engineering documents

Version	0.9
Status	In review
Date	5 October 2026
Classification	Public
Document owner	B.M.P Banneka, Product Owner

© 2026 Team Kestrel. Published under the classification shown above.

Uncontrolled when printed or exported. The controlled version is the Markdown source in the repository: docs/00-documentation-standard/ZP-DOC-00_Documentation-Standard.md

Document control

Document information

Document ID	ZP-DOC-00
Title	Documentation Standard
Version	0.9
Status	In review
Classification	Public
Document owner	B.M.P Banneka, Product Owner
Author	Team Kestrel (drafted with Claude, AI coding partner)
Conforms to	ISO/IEC/IEEE 15289:2019 (content of life-cycle information items)
Source file	docs/00-documentation-standard/ZP-DOC-00_Documentation-Standard.md
Repository	https://github.com/PraveenBanneka/zedpath

Revision history

Version	Date	Author	Description of change
0.9	5 Oct 2026	Team Kestrel	First issue, submitted for approval

Review and approval

This document takes effect when the approver signs below. Until then its status is as shown on the cover.

Role	Name	Signature	Date
Reviewer	Claude (AI coding partner), consistency review		
Approver	B.M.P Banneka, Product Owner		

Contents

1	Introduction	5
1.1	Purpose	5
1.2	Scope.....	5
1.3	Intended audience	5
1.4	Conformance.....	5
2	The document set	6
2.1	Document register	6
2.2	Folder and file layout	7
3	Identification.....	9
3.1	Document identifiers	9
3.2	Item identifiers.....	9
4	Versioning and status.....	10
4.1	Version numbers.....	10
4.2	Status lifecycle	10
5	Review and approval.....	11
5.1	Roles.....	11
5.2	Approval procedure	11
6	Change control.....	12
7	Mandatory structure.....	13
7.1	Front matter.....	13
7.2	Generated front pages.....	13
7.3	Mandatory sections	13
8	Writing rules.....	14
8.1	Requirement language.....	14
8.2	Style rules.....	14
8.3	Privacy in documents.....	14
9	Visual identity	15
9.1	Page and typography	15
9.2	Header and footer.....	15
9.3	Section numbering.....	15
10	Diagram standards.....	16
11	Traceability.....	17
12	Tooling and publication.....	18
	Appendix A: Glossary	19
	Appendix B: References.....	20

List of figures

Figure 1: Dependencies between ZedPath documents..... 7

Figure 2: Document status lifecycle..... 10

Figure 3: Traceability chain across the document set 17

List of tables

Table 1: ZedPath document register 6

Table 2: Identifier schemes for items inside documents..... 9

Table 3: Document statuses and their meaning..... 10

Table 4: Documentation roles 11

Table 5: Keywords that express obligation..... 14

Table 6: Page and typography specification 15

Table 7: Diagram notations by purpose 16

Table 8: Terms and abbreviations used in this standard..... 19

Table 9: Standards and sources referred to in this standard 20

1 Introduction

1.1 Purpose

This standard defines how every ZedPath engineering document is identified, structured, written, styled, reviewed, approved and changed. Following one standard makes the document set consistent, lets any reader find the same information in the same place, and lets every requirement be traced from the evidence that justified it to the test that proves it.

1.2 Scope

This standard applies to every document in the `docs/` folder of the ZedPath repository (ZP-DOC-00 to ZP-DOC-08 and any later additions). It does not apply to source-code comments, commit messages or the repository README, which follow the conventions in the project CLAUDE.md.

1.3 Intended audience

Authors and reviewers of ZedPath documents, and anyone assessing the quality of the engineering process (for example, buildathon judges).

1.4 Conformance

A document conforms to this standard when it meets every rule written with **shall**. Rules written with **should** are recommended practice; a document may depart from them when the reason is recorded in its revision history.

2 The document set

2.1 Document register

The ZedPath document set follows the classic engineering order: each document refines the one before it. A document **shall not** be approved before the documents it depends on are at least *In review*.

Table 1: ZedPath document register

ID	Title	Governing standard or model	Answers the question
ZP-DOC-00	Documentation Standard	ISO/IEC/IEEE 15289:2019	How do we write and control documents?
ZP-DOC-01	Vision and Scope	Wiegers and Beatty vision and scope template	Why are we building this, and what is in and out?
ZP-DOC-02	User Stories	Cohn user stories, INVEST, MoSCoW	Who needs what, and how do we know it is done?
ZP-DOC-03	Software Requirements Specification	ISO/IEC/IEEE 29148:2018	What exactly shall the system do, and how well?
ZP-DOC-04	Use Case Model	UML 2.5.1; Cockburn fully dressed use cases	How do actors achieve their goals, step by step?
ZP-DOC-05	Data Design	Enhanced ER (Elmasri and Navathe); relational normalisation	What data exists, how is it related and stored?
ZP-DOC-06	Software Architecture	ISO/IEC/IEEE 42010:2022; C4 model; arc42; ADRs	How is the system structured and deployed, and why?
ZP-DOC-07	UI/UX Specification	Material Design 3; WCAG 2.2 AA	What does the user see and how do they move through it?
ZP-DOC-08	Test Plan	ISO/IEC/IEEE 29119-3:2021	How do we prove the system meets its requirements?

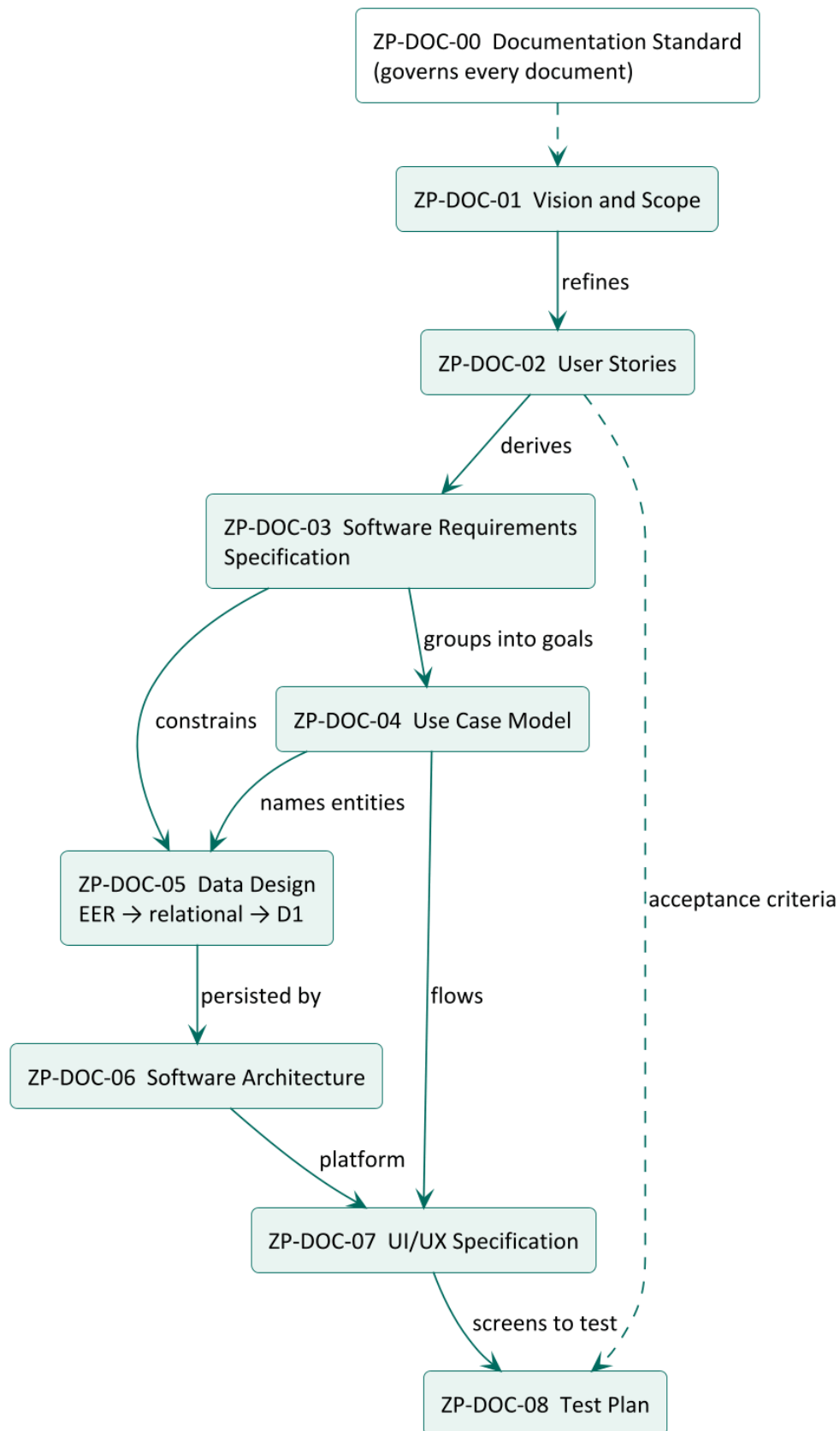


Figure 1: Dependencies between ZedPath documents

2.2 Folder and file layout

- Each document **shall** live in its own folder `docs/NN-kebab-title/`, where *NN* is its two-digit number.

- The source file **shall** be named `ZP-DOC-NN_Title-In-Words.md`.
- Generated files **shall** sit beside the source with the same base name: `.docx` and `.pdf`.
- Diagram sources **shall** live in `diagrams/` inside the document folder, as `.puml` files, with the rendered `.png` beside them.

3 Identification

3.1 Document identifiers

Every document has a permanent identifier **ZP-DOC-NN**. Identifiers are never reused, even if a document is withdrawn.

3.2 Item identifiers

Items inside documents carry identifiers so they can be referenced and traced. Identifiers **shall** be unique across the document set and **shall** never be renumbered or reused after approval; a deleted item keeps its number and is marked *Deleted*.

Table 2: Identifier schemes for items inside documents

Item	Pattern	Example	Defined in
Evidence source	<i>En, Tn</i>	E4, T1	ZP-DOC-02
Persona	<i>Pn</i>	P1	ZP-DOC-02
Epic	<i>EPn</i>	EP3	ZP-DOC-02
User story	<i>US-enn</i> (epic digit + number)	US-302	ZP-DOC-02
Acceptance criterion	<i>AC-enn.k</i>	AC-302.2	ZP-DOC-02
Business objective	<i>BO-n</i>	BO-1	ZP-DOC-01
Feature	<i>FE-n</i>	FE-4	ZP-DOC-01
Business rule	<i>BR-nnn</i>	BR-012	ZP-DOC-03
Functional requirement	<i>FR-nnn</i>	FR-045	ZP-DOC-03
Non-functional requirement	<i>NFR-nnn</i>	NFR-007	ZP-DOC-03
Use case	<i>UC-nn</i>	UC-03	ZP-DOC-04
Entity or table	singular noun in PascalCase (conceptual); snake_case (physical)	Offering; offering	ZP-DOC-05
Architecture decision	<i>ADR-nnn</i>	ADR-001	ZP-DOC-06
Risk	<i>RK-nn</i>	RK-04	ZP-DOC-01
Test case	<i>TC-nnn</i>	TC-118	ZP-DOC-08

4 Versioning and status

4.1 Version numbers

- **0.1 to 0.8**: working drafts.
- **0.9**: complete draft submitted for review.
- **1.0**: first approved version.
- **1.x**: approved revisions that do not change scope (corrections, clarifications, added detail).
- **2.0, 3.0...**: approved revisions that change scope or invalidate earlier commitments.

4.2 Status lifecycle

Every document **shall** show one of these statuses on its cover, in its header and in its document control page.

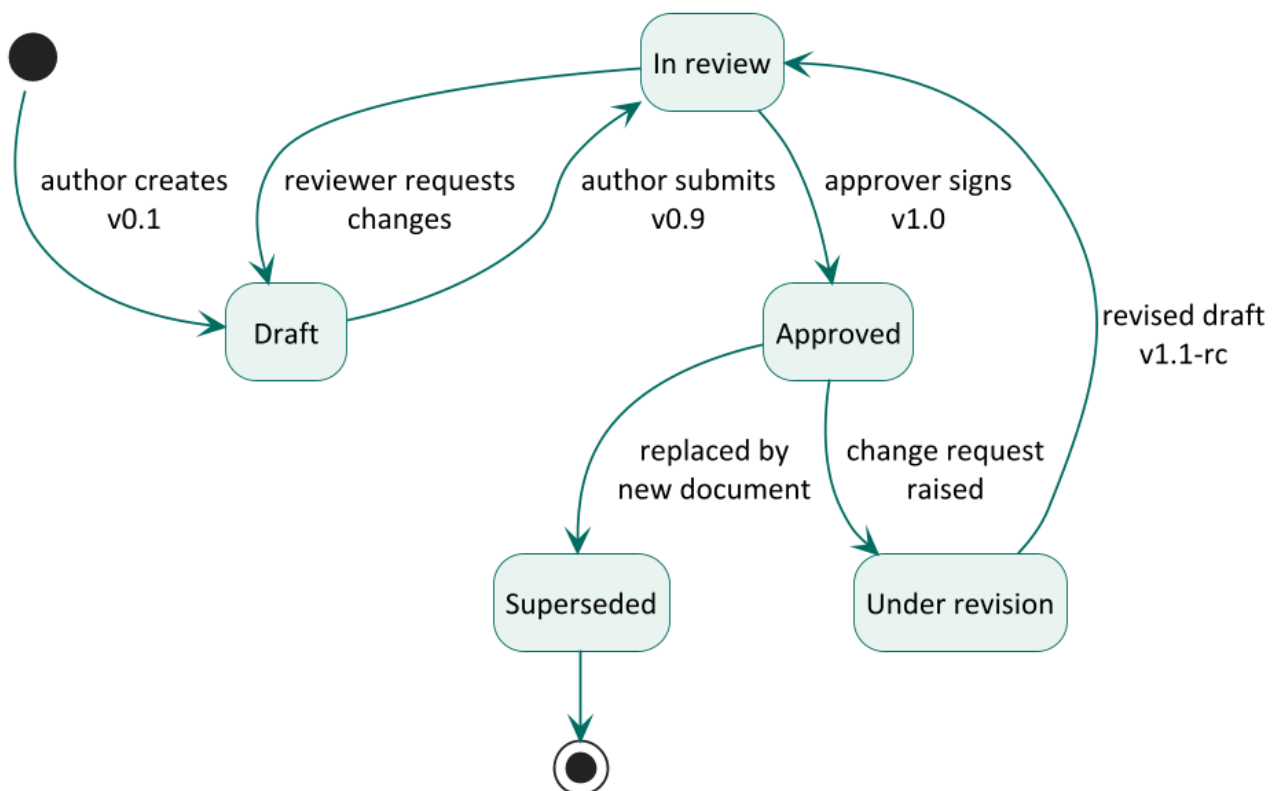


Figure 2: Document status lifecycle

Table 3: Document statuses and their meaning

Status	Meaning	Who changes it
Draft	Being written; content may change without notice	Author
In review	Complete; awaiting review comments and approval	Author submits
Approved	Signed off by the approver; the baseline for later work	Approver
Under revision	An approved document is being changed; the last approved version stays in force	Author
Superseded	Replaced by another document named in its revision history	Approver

5 Review and approval

5.1 Roles

Table 4: Documentation roles

Role	Responsibility	Held by
Document owner	Accountable for the content being correct and current	B.M.P Banneka, Product Owner
Author	Writes and revises the document	Team Kestrel (drafted with Claude)
Reviewer	Checks consistency, completeness and traceability	Claude (AI coding partner) and the owner
Approver	Accepts the document as the baseline	B.M.P Banneka, Product Owner

5.2 Approval procedure

1. The author sets the version to 0.9 and the status to *In review*, and commits.
2. Reviewers record findings; the author resolves each one and adds a revision-history row.
3. The approver signs the approval table (name and date), sets version 1.0 and status *Approved*.
4. The approval commit is tagged `ZP-DOC-NN-v1.0` and noted in `WORKLOG.md`.

AI-generated content **shall** be reviewed by the product owner before approval. An AI reviewer may check consistency and traceability, but cannot approve.

6 Change control

- The Markdown source is the **controlled** version. Generated `.docx` and `.pdf` files **shall not** be edited by hand; they are rebuilt with `node docs/_build/build.ts`.
- Every new version **shall** add a revision-history row stating what changed and why.
- A change to an approved item **shall** be checked against everything that traces from it (see the traceability rules below), and those documents updated in the same commit or flagged in `WORKLOG.md`.
- Every change is committed to the public repository; Git history is the audit trail.

7 Mandatory structure

7.1 Front matter

Every source file **shall** begin with a front-matter block containing: `id`, `title`, `subtitle`, `version`, `date`, `status`, `classification`, `owner`, `author`, `approver`, `standard` (the governing standard), at least one `revision` line, and optionally `reviewer` lines.

7.2 Generated front pages

The build tool generates, in this order: the **cover page**; the **document control** page (document information, revision history, review and approval table); the **contents**; the **list of figures** and **list of tables** when the document has any.

7.3 Mandatory sections

Every document **shall** contain:

1. **Introduction**, with at least Purpose, Scope and Intended audience. Documents that sit in the chain **shall** also state their relationship to other documents.
2. **Body** sections required by the governing standard named in the register.
3. **Appendix: Glossary** of terms and abbreviations used.
4. **Appendix: References** to every external source cited.

8 Writing rules

8.1 Requirement language

Table 5: Keywords that express obligation

Keyword	Meaning
shall	Mandatory; the system or document is non-conforming without it
should	Recommended; departures need a recorded reason
may	Optional
will	A statement of fact or intent, not a requirement

8.2 Style rules

- Write in plain British English, in the active voice, in short sentences.
- One requirement per statement. Each requirement **shall** be verifiable.
- Avoid unverifiable words: *user-friendly, fast, easy, flexible, robust, etc., and so on, as appropriate*. Replace them with numbers and units (for example, "within 2 seconds on a 3G profile").
- Write dates in prose as *5 October 2026*; in data, tables of values and identifiers, use *ISO 8601 2026-10-05*.
- Refer to sections in words: "Section 4.2". The section-sign symbol **shall not** be used.
- Define every abbreviation on first use and in the glossary.
- Sinhala or Tamil terms are written in English with the original in brackets on first use where it helps.
- Quote evidence verbatim and attribute it to its code (for example, E4).
- Do not refer to a table or figure by number in running text; describe it ("the table below"), because numbers are assigned automatically.

8.3 Privacy in documents

Documents are public. They **shall not** contain personal data. Interviewees are referred to only by codes (E1 to E6). Account identifiers, keys and tokens **shall never** appear in a document.

9 Visual identity

9.1 Page and typography

Table 6: Page and typography specification

Element	Specification
Page	A4 portrait; margins 2.0 cm (top 2.35 cm, to clear the header); header and footer 1.0 cm from the edge
Body text	Calibri 10.5 pt, line spacing 1.1, colour #1B1F1E
Heading 1	Calibri 17 pt bold, ZedPath green #006B5F, rule below, starts a new page
Heading 2	Calibri 13.5 pt bold, #1B1F1E
Heading 3	Calibri 11.5 pt bold, dark green #00504A
Captions	Calibri 9 pt italic, #55605D, numbered automatically: "Table n:" above tables, "Figure n:" below figures
Code	Consolas 8.5 pt on a light grey panel
Tables	Header row green #006B5F with white bold text, alternate rows tinted #F3F8F6, 0.5 pt grey rules
Call-outs	Tinted panel with a green bar on the left; used for notes and the narrative of user stories

9.2 Header and footer

- **Header:** "ZedPath | *document title*" on the left; "*document ID* · *vversion* · *status*" on the right.
- **Footer:** "Classification: Public" on the left; "Uncontrolled when printed" in the centre; "Page *n* of *N*" on the right.
- The cover page carries no header or footer.

9.3 Section numbering

Headings are numbered automatically: 1, 1.1, 1.1.1. Appendices are lettered: Appendix A, A.1. Front pages (document control, contents and lists) are not numbered and do not appear in the contents.

10 Diagram standards

Diagrams are written as code (PlantUML) so they can be reviewed and versioned like text. Each diagram **shall** use the notation below for its kind, **shall** have a caption, and **shall** be readable when printed in greyscale.

Table 7: Diagram notations by purpose

Purpose	Notation	Used in
Story map	Mind map	ZP-DOC-02
System context	C4 model, level 1 (System Context)	ZP-DOC-01, ZP-DOC-06
Use cases	UML 2.5.1 use case diagram	ZP-DOC-04
Behaviour and flows	UML activity and sequence diagrams	ZP-DOC-04, ZP-DOC-06
Life cycles	UML state machine diagram	ZP-DOC-00, ZP-DOC-05
Conceptual data model	Enhanced Entity-Relationship (EER), Chen notation as in Elmasri and Navathe: rectangles for entities, diamonds for relationships, ovals for attributes, underlined keys, double lines for weak entities and total participation, circles for specialisation with <i>d</i> (disjoint) or <i>o</i> (overlapping)	ZP-DOC-05
Logical data model	Relational schema diagram, crow's foot (Information Engineering) notation	ZP-DOC-05
Containers and deployment	C4 level 2 (Container) and UML deployment diagram	ZP-DOC-06
Screen flow	Navigation map	ZP-DOC-07

11 Traceability

Every item **shall** trace to its parent, so that any requirement can be followed from the evidence that justified it to the test that proves it, and any test back to its reason.

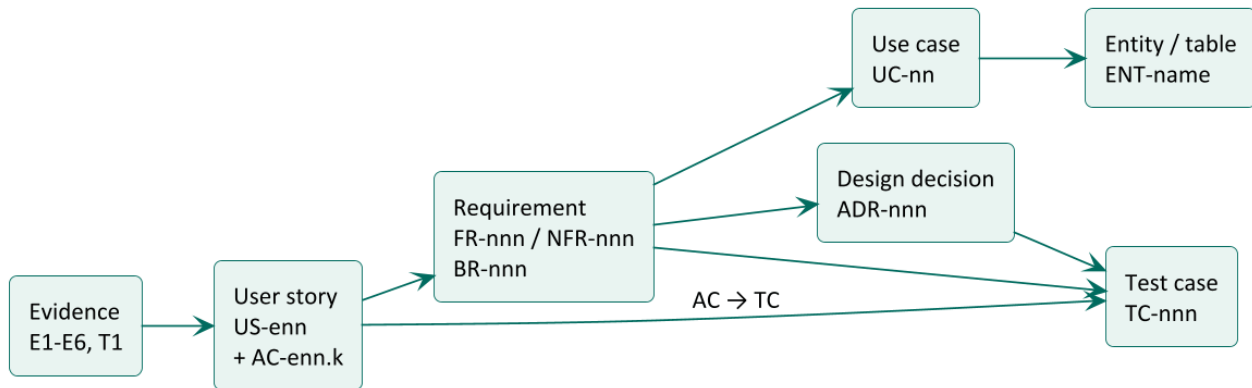


Figure 3: Traceability chain across the document set

- Each document **shall** contain a traceability table to its parent document.
- ZP-DOC-08 **shall** contain the full matrix from requirement to test case.
- An item with no parent is either unjustified (remove it) or reveals missing evidence (record it).

12 Tooling and publication

- Sources are Markdown with front matter; diagrams are PlantUML; the build tool (docs/_build/build.ts, TypeScript) produces .docx with docx-js and .pdf through Microsoft Word, which also refreshes the contents, the lists of figures and tables, and page numbers.
- Generated files are committed, so readers without the tools can open every document.
- The repository is public, so every document is classified **Public**.
- Official third-party documents (for example UGC handbooks and cut-off tables) are cited but **shall not** be redistributed in the repository.

Appendix A: Glossary

Table 8: Terms and abbreviations used in this standard

Term	Meaning
ADR	Architecture Decision Record: a short record of one significant design decision and its reasons
Baseline	An approved version that later work builds on
C4 model	A model for software architecture diagrams at four levels: context, container, component, code
EER	Enhanced Entity-Relationship model: ER plus specialisation, generalisation and categories
PlantUML	A tool that draws diagrams from text descriptions
Traceability	The ability to follow an item to its origin and to everything derived from it
UML	Unified Modeling Language, maintained by the Object Management Group

Appendix B: References

Table 9: Standards and sources referred to in this standard

Ref	Document
[1]	ISO/IEC/IEEE 15289:2019, <i>Systems and software engineering: Content of life-cycle information items (documentation)</i>
[2]	ISO/IEC/IEEE 29148:2018, <i>Systems and software engineering: Life cycle processes, Requirements engineering</i>
[3]	ISO/IEC/IEEE 42010:2022, <i>Software, systems and enterprise: Architecture description</i>
[4]	ISO/IEC/IEEE 29119-3:2021, <i>Software and systems engineering: Software testing, Part 3: Test documentation</i>
[5]	K. Wiegers and J. Beatty, <i>Software Requirements</i> , 3rd edition, Microsoft Press, 2013 (vision and scope template)
[6]	R. Elmasri and S. Navathe, <i>Fundamentals of Database Systems</i> , 7th edition, Pearson, 2016 (EER notation)
[7]	Object Management Group, <i>Unified Modeling Language</i> , version 2.5.1, 2017
[8]	A. Cockburn, <i>Writing Effective Use Cases</i> , Addison-Wesley, 2001
[9]	S. Brown, <i>The C4 model for visualising software architecture</i> , c4model.com
[10]	M. Nygard, <i>Documenting Architecture Decisions</i> , 2011
[11]	W3C, <i>Web Content Accessibility Guidelines (WCAG) 2.2</i> , 2023
[12]	Google, <i>Material Design 3</i> , m3.material.io